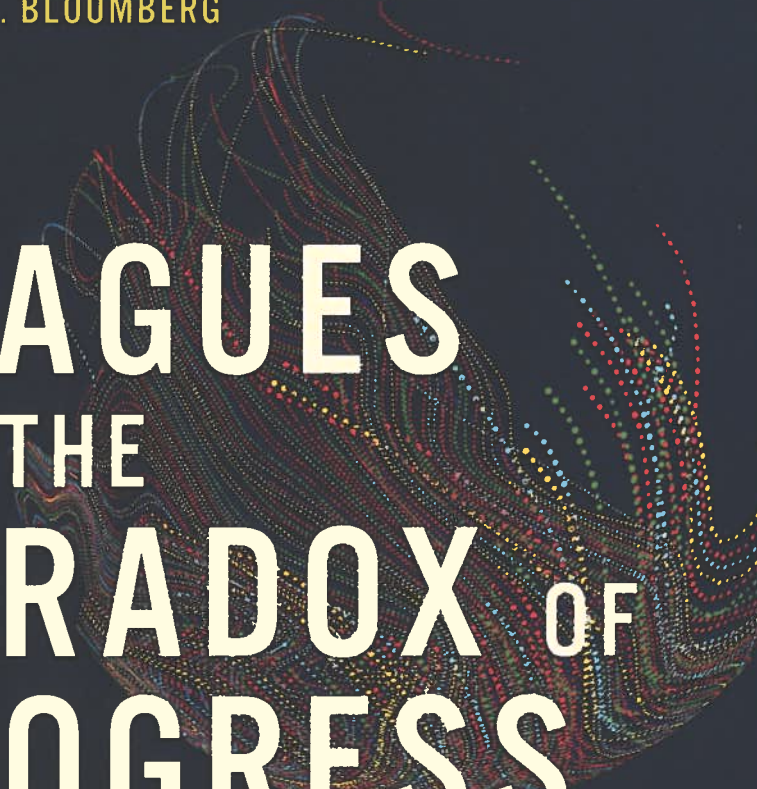


"BOLLYKY'S BOOK CAN HELP TO SAVE MANY LIVES."
—MICHAEL R. BLOOMBERG



PLAGUES AND THE PARADOX OF PROGRESS

WHY THE WORLD IS GETTING HEALTHIER
IN **WORRISOME** WAYS

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We are in an era of great and welcome progress in the global fight against infectious disease. That fight must continue and we should continue to invest in it. But we must also recognize that making this progress against plagues and parasites has brought about new and unintended challenges, as important progress often does. For us to seize the opportunities and prosperity that came with health improvements in the past, these emerging challenges must be recognized and overcome. Failing to do so may undermine or reverse hard-fought improvements in health and living standards that have been achieved in recent decades.

This book is not just about the role of plagues in the sweep of history, but the countless lives that were lost, one by one, to those diseases, and the precious opportunities to be gained when those lives are saved in an environment where individuals can take advantage of those opportunities. The aim of this book is to accomplish what William McNeill's classic book *Plagues and Peoples* has done: to move its readers to see what is happening in the world differently and to be motivated to do something about it.

Introduction

Infectious disease which antedated the emergence of humankind will last as long as humanity itself, and will surely remain, as it has been hitherto, one of the fundamental parameters and determinants of human history.

—William H. McNeill, *Plagues and Peoples*

No other development over the last hundred years—not world wars, not the internet, not the spread of democracy—has had as widespread and transformative an impact on the human experience as the decline in infectious diseases. This book is about the paradox of that progress in the long fight against parasites, viruses, and other plagues: the world is getting better in ways that should make us worry.

The slow decline in the death and disability from infectious diseases began 250 years ago with the Enlightenment and the Industrial Revolution in Northern Europe and spread to the United States and other countries with deep European roots. After World War II, this trend began to accelerate and extend globally and, in recent years, has reached even the poorest countries. The consequences of this expanded control of infectious disease are hard to overstate.

As recently as sixty years ago, most developing nations were still subject to the cycle of low life expectancy and high birth rates that have characterized humankind for millennia.¹ In 1950, one out of five children still perished before their fifth birthday in nearly one hundred countries, including almost every nation in sub-Saharan Africa, South Asia, and Southeast Asia. The average life expectancy of a baby born in these nations was 42 years. With so many dying young, few children received a formal education, and most grew up to live lives similar to those of their parents and the generations before them: in rural settings and engaged in subsistence farming. There

was a thirty-three-year difference between the average life expectancy in Northern Europe (70 years old) and sub-Saharan Africa (37 years old).

Lower-income nations have significantly narrowed these gaps since the 1950s. By 2015, the average baby born in a developing country was expected to live to 70. The number of countries where one out of five children perish under the age of five is now zero. With fewer children and adolescents dying, investment in education in many lower-income nations has dramatically improved. The average adult in Bangladesh, Haiti, and Zambia now has more years of schooling than the average adult in France or Italy did in 1960.² Agriculture no longer employs the majority of people in most regions of the world. Just over half of those living in developing countries now reside in a city. The total number of people in lower-income nations has expanded from 1.7 billion in 1950 to 6.1 billion in 2015. Despite this population increase, per-capita incomes in developing nations have grown and extreme poverty has declined, driven by increased global trade, Asian manufacturing, and rising commodity prices. By 2015, the gap in life expectancy between Northern Europe and sub-Saharan Africa had shrunk by ten years.

All this progress that has been achieved against plagues and parasites is one of the great achievements of humankind, but it is also spurring new and daunting challenges. When richer countries took on the scourge of infectious diseases, few advanced medicines were available. Their tools, therefore, included social controls, like quarantines, and investments in effective water and sewage systems. These tools required or led to more responsive governments and capable health-care systems, which in turn helped advance the prosperity of those nations.

In contrast, the recent gains in infectious disease control in many lower-income nations have been more dependent on international aid and effective medical technologies.³ As a result, the decline in infectious disease in many (but by no means all) lower-income nations has not been accompanied by the same improvements in personal wealth, health care, responsive governments, and employment opportunities that accompanied those changes in today's wealthier nations. In the absence of those broader improvements, many emerging economies are struggling to cope with the demographic changes that have occurred with the decline of infectious diseases—unprecedented urbanization and a rapid rise in the number of working-age adults. At the same time, the world is changing, with wealthy nations adopting trade and immigration policies that are much

less accommodating of these demographic changes happening in poorer nations. Climate change is making it harder for farmers and pastoralists in many regions to earn a living, adding to the pressure for them to migrate. If donors and international institutions remain slow to respond to these emerging challenges, there could be dire consequences that resound through the world's economic, political, and security landscape.

To explain the cause for concern, this book must convince you of three insights: first, that infectious diseases are indeed declining; second, that the threat of infectious diseases and the efforts to overcome that threat have played and will continue to play an important role in shaping the modern world; and third, that the way infectious diseases are declining in recent decades is different from how they have declined in the past, and that difference, when combined with broader global changes, is producing deeply worrisome consequences for the future. I outline these arguments briefly in this introduction, but we will explore them in depth in the pages that follow.

The Age-Old Balance between Host and Parasite

The notion that infectious diseases are in decline may be hard to believe. There are still 36.7 million people worldwide living with HIV/AIDS. More than six million people were newly infected with tuberculosis in 2016. The World Health Organization estimates that roughly 185,000 people die annually from neglected tropical diseases.⁴ Outbreaks of exotic parasites, bacterial blights, and obscure tropical viruses dominate the news. Deservedly, much attention has been given to the lives lost and children disabled in high-profile epidemics of influenza, severe acute respiratory syndrome (SARS), and the Ebola and Zika viruses. Reports of smaller outbreaks of well-known infections and encounters with new diseases continue to occur regularly.⁵ People and plagues, McNeill wrote, exist in "an age-old balance between host and parasite"; and those infectious diseases will remain "a permanent feature of human (and all multi-cellular forms of) life."⁶

New and previously unknown infectious diseases are likely to remain an ever-present threat to human life. Viruses, such as Ebola or avian influenza, continue to jump from animals to humans, evolve, and intermingle in ways that make them deadlier and more prone to spread. Overuse of existing drugs and underinvestment in new ones are producing drug-resistant strains of fungi, protozoa, tuberculosis, and other bacteria, making routine medical

care more dangerous. These emerging and treatment-resistant pathogens easily cross national boundaries, given increases in global trade, faster travel, and rising global temperatures (resulting in warmer, more virus-friendly climates).⁷ The risk of another great pandemic of infectious disease—similar to the Black Death of bubonic plague in the fourteenth century, cholera in the nineteenth century, influenza in 1918–19, or HIV/AIDS—is not behind us.

Given our increasingly intertwined world, these threats are still worthy of serious concern, even if they are not entirely new. Infectious disease has accompanied trade since at least 430 BCE, when a plague brought by infected rats aboard a trading ship killed a third of the population of Athens. Cholera spread in the nineteenth century via railways across North America and on steamships to the United Kingdom and the rest of Europe.⁸ Resistant strains of bacteria predated the discovery of the first antibiotics.⁹ Climate change allowed malaria to expand out of Africa centuries ago and likely influenced all three major pandemics of bubonic plague, including the Black Death of the Middle Ages.¹⁰ New risks such as synthetic, man-made viruses may yet emerge, but for now, the differences between the contribution of globalization, climate change, and drug resistance to today's infectious disease threats and those of the past are largely matters of degree, not kind. The long-standing process of managing these threats remains deeply important.

But that ongoing threat should not blind us to what has changed. In the last few decades—for the first time in recorded human history—infectious diseases have declined significantly, even in the poorest populations. According to the University of Washington's Institute for Health Metrics and Evaluation, a leading source of data on global health, infectious diseases now account for less than 8 percent of the death and disability in most regions of the world, including many lower-income and tropical regions such as the Middle East and North Africa, and Latin America and the Caribbean. There are only two exceptions. The first is South Asia, which includes India, Pakistan, and more impoverished nations like Bangladesh and Nepal; infectious diseases are responsible for roughly one-fifth of the health burden in that region. The second is sub-Saharan Africa, where plagues and parasites stopped being the cause of the majority of death and disability in 2011. Four years later, that rate fell to less than 44 percent in this poorest of geographic regions.

The long-term declines in infectious diseases have proven durable, even amid outbreaks and epidemics of new and reemerging infections. HIV/AIDS

has exacted a terrible toll, disproportionately in sub-Saharan Africa. At its peak in 2005, the disease killed 1.8 million people annually. With the efforts of the US government and other donors and partners to expand access to antiretroviral treatment and prevent HIV/AIDS transmission from mothers to their infants, death rates from the disease dropped 42 percent in a decade. Even in countries with the highest rates of HIV/AIDS infection, such as Botswana, average life expectancy rates have bounced back and now exceed pre-epidemic levels.

The recent progress and resilience against HIV/AIDS and other infectious diseases reflect the tireless, coordinated efforts of many. The relative success of their efforts was not a foregone conclusion, and there is much more left to do, especially in the poorest countries. Population growth, urbanization, and the decline of endemic infectious diseases in many countries have increased the risks of future outbreaks and epidemics by bringing larger numbers of people without prior exposure to infections into closer contact with each other. The risks of emerging infections, antibiotic resistance, and bioterrorism remain important, and preparing for future pandemics requires sustained global investment.

But the continued threat of microbes to humankind should not obscure the overall decline in rates of infectious disease any more than a terrible snowstorm would disprove the otherwise steady rise in global temperatures. Indeed, as with climate change and extreme weather events, there may be ties between the falling global rates of everyday plagues and parasites and the heightened risk of occasional infectious disease outbreaks. The decline in infectious diseases is unprecedented and sustained, and it is transforming lower-income countries. Most of that transformation has happened in just the last few decades. The effects extend beyond longer lives and less suffering, to the growth and the geography of cities and to shifts in global economic power and patterns of human migration. Not all these changes have been positive.

Determinants of History, Agents of Human Tragedy

To understand how and why the decline of infectious diseases is reshaping the world, for good and for ill, we need to return to McNeill. He argued in *Plagues and Peoples* that infectious diseases have been “one of the fundamental parameters and determinants of human history.” Subsequent

scholars have joined McNeill in the observation that the power of microbes has often extended beyond their death toll.¹¹ Three traits account for the outsized influence of infectious diseases over human history.

First, infectious diseases, by definition, can spread. That transmission can occur from person to person or indirectly through an insect or animal intermediary such as a mosquito or rat. The risk of transmission is greatest in settlements and with activities that bring large numbers of people into close contact. For this reason, encounters with infectious disease have played a key role in the evolution of cities, the expansion of trading routes, the conduct of war, and participation in pilgrimages. Again, it is hard to improve on Zinsser:

Swords and lances, arrows, machine guns, and even high explosives have had less power over the fates of the nations than the typhus louse, the plague flea, and the yellow fever mosquito. Civilizations have retreated from the plasmodium of malaria, and armies have crumbled into rabble under the onslaught of cholera spirilla, or of dysentery and typhoid bacilli. Huge areas have been devastated by the trypanosome that travels on the wings of the tsetse fly, and generations have been harassed by the syphilis of a courtier. War and conquest and that herd existence which is an accompaniment of what we call civilization have merely set the stage for these most powerful agents of human tragedy.¹²

Second, infectious diseases disproportionately affect previously unexposed people. When plagues and parasites do not kill, survivors are often left with life-long immunity to a reoccurrence of the disease. Many histories of infectious disease have focused on ways in which populations without that prior immunity have borne the brunt of the microbes that accompanied invading armies, trade caravans, and explorers. The risk that plagues and parasites pose for people without prior immunity is also why many infectious diseases disproportionately affect children. Infectious diseases have kept rates of child mortality high throughout most of human history, which has both led mothers to have more children in order to compensate for those who might be lost and undercut the incentive for families to invest in their young. Those consequences, in turn, have had profound implications for the role of women in societies, education rates, and the composition of labor forces.

The third reason for the outsized role of infectious disease in human history is that their prevention and control depends on the cooperation of people and governments. Individuals, households, communities, and

nations cannot isolate themselves from the risk of infectious disease for long, and can do so only at great cost.¹³ Measures to prevent and control infectious disease can succeed and be sustained only if our neighbors, other communities, and national governments undertake those measures too. The Nobel Laureate Joshua Lederberg once aptly wrote: "The microbe that felled one child in a distant continent yesterday can reach yours today and seed a global pandemic tomorrow."¹⁴

Other threats to human health, such as tobacco products, illicit drugs, and air pollution, can also cross national borders and require a collective response to be effective. But infectious diseases predate these other concerns as the first global problem that nation-states realized could not be solved without international cooperation. The words that the English political theorist Leonard S. Woolf wrote in 1916 still ring true: "The conflict fought by the theory of national independence, isolation, and national interests against the facts of international life and international interests has nowhere shown itself more persistently and clearly than in the struggle of human beings against the scourges of cholera, plague, and other epidemic diseases."¹⁵ Infectious diseases have been a particularly good motivator of collective action because of the disproportionate threat they pose to populations without prior exposure, the dread that these diseases inspire, and the economic losses that may result when epidemics disrupt international trade and travel.¹⁶

The Different Paths to Progress

Infectious diseases forced governments to assume greater administrative power and authority over the practices of their citizens and commercial enterprises. The understanding that some diseases were contagious dates back to the Bible, which includes detailed rules for isolating lepers in their homes.¹⁷ With the first major Black Death epidemic in the fourteenth century, local authorities in Venice and Florence applied that Old Testament practice of forced isolation to persons in transit.¹⁸ The word *quarantine* derives from *quaranta giorni*, which means "forty days" in Italian. Renaissance-era governments adopted other public health innovations to cope with epidemics as well: requiring civil registration of citizens and forming the first municipal boards of health. These measures represented a significant expansion of state authority and were copied by other countries struggling to cope with outbreaks of plague and parasites.¹⁹

When the threat of epidemics involved cholera and other microbial threats where quarantine was ineffective, governments adapted, expanding again on examples from the past. Jewish, Hindu, and Islamic laws include rules on bodily cleanliness and the handling of food and water. The Romans built an elaborate network of aqueducts, underground channels, and pipes to drain the surrounding swamps, which stemmed the tide of malaria and provided sanitation and drinking water.²⁰ Under pressure from social reformers and angry citizen mobs, governments of wealthy countries in the nineteenth century constructed water and waste-management systems, adopted housing codes and food regulations, promoted personal hygiene, and entered into the first international health treaties to contain epidemic threats.²¹

Most of the gains in infectious disease control in higher-income nations occurred before the invention of effective medical treatments for those diseases. The precise mix of measures undertaken varied by disease and by country, as did the rate of the progress.²² The path to progress against pestilence was not linear; countries suffered reversals and important innovations, such as the germ theory of disease, took decades to be widely adopted.²³ But many of the nations that experienced significant, long-term declines in plagues and parasites shared common themes: increased industrialization and a rising middle class, the advent of trade unions and better women's education, and the establishment of public health laws and more responsive government institutions.²⁴ In some cases, control of infectious disease was largely the result of changes such as rising incomes and increased access to education. In others, infectious disease control spurred investment in sanitation, housing laws, and municipal infrastructure that then had broader societal and economic benefits for now-wealthy nations. In these ways, the slow process of infectious disease control both required and allowed many governments to improve their functioning. That commitment cannot flag; where developed countries have failed to sustain their investments in controlling plagues and parasites, they have suffered resurgences of diseases such as tuberculosis.²⁵

The recent decline in plagues, viruses, and parasites in poorer countries is just as important as those past health improvements were in now-wealthy nations. But the efforts to control infectious disease threats in many lower-income nations are not driving the same social reform and investment in municipal, national, and international governance as those efforts did decades ago. The heavy burden of plagues, viruses, and parasites is being overcome in the cities of many lower-income nations without effective

housing laws, adequate municipal water and sewage systems, and the economic infrastructure needed to attract factories that can put migrants to work. The particular risk that infectious diseases pose to previously unexposed peoples, particularly infants and children, is being surmounted without commensurate improvements in public health and in the status of women.

Why Worry in the Age of Miracles?

Michael Elliott, the late journalist and former CEO of the ONE Campaign, rightly called this era in global health "the age of miracles."²⁶ Better infectious disease control in lower-income nations has saved millions of children and infants who otherwise would have perished or lived a life of infirmity.

In too many countries, however, too little is being done to ensure that the children and infants surviving to adulthood find employment opportunities and adequate health systems to accommodate their needs as adults. The World Bank estimates that the working-age population (more than 15 years old) in developing countries will increase by 2.1 billion by 2050. Unless current national employment rates improve, that will mean nearly 900 million more young adults without work.²⁷ Rates of chronic diseases that are most associated with affluent nations, such as heart disease, cancer, and diabetes, are rising fast in lower-income countries without capable health-care systems to prevent and manage these diseases. The growth of cities in many poorer nations is historically unprecedented, far outpacing their urban economies and infrastructure. For the first time in history, the existence of large cities is not necessarily indicative of prosperity and good governance. Congested metropolises and the lack of formal employment for rising youth populations threaten to breed instability. Climate change and increased hostility to trade and immigration in wealthy nations are exacerbating the situation by leaving poorer nations with fewer avenues for expanding job opportunities and restricting the ability of their citizens to move elsewhere. It is not hard to envision a scenario where the next several decades are less peaceful than the last.

A Worrisome Future Is Not Inevitable

The story of the decline in infectious diseases threatens to be as consequential as the history of their rise. It deserves to be recognized and treated as such. But a dire future is not inevitable. This book is not a treatise on

overpopulation in developing countries or how the world would have been better off forgoing investment in global health. The planet does not thrive when the sickest are permitted to die off, but rather when they are able to improve their lives. Efforts to reduce infectious disease, early death, and unnecessary suffering in lower-income countries should continue. National governments, donors, and intergovernmental institutions should do everything possible to accelerate that process.

More people need not mean more people who are poor. Reductions in infectious disease can accelerate economic growth and better governance when accompanied by improvements in educational quality, infrastructure, and formal employment opportunities. Countries as diverse as Brazil, Iran, and Saudi Arabia have achieved fertility declines without resorting to draconian forms of population control. Despite long-standing fears and predictions to the contrary, innovations in agricultural production in the twentieth century have continued to enable it to exceed the rate of population growth.

But rejection of neo-Malthusian fears about hordes of people breeding in poorer nations does not mean we should ignore the new challenges emerging as a result of the progress made in the fight against infectious diseases. We should not assume that poor countries can or will follow the same path as wealthier nations.

Once, it was the more advanced nations of the world that were more disease-ridden. The disproportionate number of deaths from plagues and parasites in impoverished tropical and subtropical nations is a recent development in human history. Chapter 1 tells the story of how that changed, the link between the history of infectious diseases and agriculture, trade, urbanization, and humankind's other civilizing habits, and the long, slow route that wealthy nations took to better health. This chapter opens in South Africa in 2003, amid a raging HIV/AIDS epidemic. It is the moment when the gap between the incidence of infectious disease in the poorest nations and the wealthiest ones was arguably the largest in recorded human history.

The next four chapters describe the way that plagues and parasites have declined more recently in poorer nations, the different paths that this progress has taken, and the consequences. Each chapter focuses on a theme and uses case studies of particular diseases and countries to explore that theme.

Chapter 2 starts with diseases of conquest and colony: smallpox and malaria. The theme of this chapter is the link between infectious disease,

foreign aid, and the state. It examines the military and colonial origins of the global disease eradication campaigns in the 1950s and their influence on the disease-specific, technology-driven strategies used to reduce the plagues and parasites in poorer nations. This approach has proved remarkably successful in addressing the targeted infectious diseases, but has not been accompanied by the broad improvements in responsive and more capable governments that occurred with the control of everyday infectious diseases in many wealthy nations. When combined with rapid population growth, urbanization, and more adults, the underdeveloped health-care systems in poor countries are straining to cope with the threat of new and emerging infections like Ebola and an unprecedented increase in noncommunicable diseases.

Chapter 3 turns to the diseases of childhood—measles and other respiratory diseases—that disproportionately affect children once they become endemic. The theme of this chapter is the link between infectious disease and economies. It starts with an audacious campaign launched in 1982 to extend childhood immunizations to the poorest regions of the world in just eight years. It then considers how the decline in mortality from childhood diseases helped transform the economic fortunes of some formerly poor countries in the space of just a few decades. This chapter also explains why the economic opportunity afforded by that decline in infectious disease is fleeting and potentially perilous. That story features China and Kenya.

Chapter 4 considers the diseases of settlement—tuberculosis, and cholera and other intestinal diseases—that rose to prominence in the first industrializing cities of Europe and America. The theme of this chapter is the link between infectious disease and urbanization. Frugal innovations to tackle these diseases have helped shift the geography and growth of mega-cities, a phenomenon once reserved for wealthy nations, but now found most often in lower-income countries. This chapter begins on the busy streets of Dhaka, Bangladesh, the most densely populated city on earth.

Chapter 5 addresses the diseases of place—meningitis and other neglected infectious and parasitic diseases—that occur almost exclusively in the poorest and most ecologically challenged regions of the world. These diseases have been the targets of heroic international health campaigns in the last fifteen years and are starting to decline. The infants and children saved by these recent health improvements are now starting to reach adulthood and to do what many of the beneficiaries of infectious disease control have

always done: emigrate. The theme of this chapter is infectious disease and migration, and it focuses on Niger and Ireland.

The sixth and concluding chapter of this book opens with the story of William H. Stewart and one of the most famous, widely used, and erroneously attributed quotes in modern medicine. This chapter explains how the statement Stewart actually made is a useful frame for thinking about policy recommendations to build a world that continues to get better, just in less worrisome ways.

There is a future for lower-income nations that includes more inclusive economic growth, smart investments in health-care systems and education, freer trade, continued advances in technology, sustainable agricultural production, and better governance. Only the governments of those countries and the people who live there can realize that potential future. But foreign aid agencies, philanthropists, and the private sector can do much more to help those countries sustainably tackle their crushing demographic challenges. This book lays out that agenda, but to adopt and implement those solutions effectively we need a greater understanding of the nature and basic history of the problem and where and why it is occurring. That is the task of the next several chapters of this book.

1 How the World Starts Getting Better

Where are the drugs? That's where they are, the drugs are where the disease is not. And where is the disease? The disease is where the drugs are not.

—Dr. Peter Mugenyi, Durban AIDS Conference, 2000

"I have an idea."

It was my first day of work at the AIDS Law Project, a nongovernmental organization based in Johannesburg, South Africa. I was fresh out of law school, eager to contribute. So when my new boss asked at the weekly staff meeting if anyone knew how to safely get a syringe of the drug Nevirapine to a woman about to give birth in a rural part of the country, I volunteered.

The intended recipient of that syringe, S,¹ was an expecting first-time mother and one of the four million South Africans living with HIV/AIDS in 2001.² One out of every four women at a public antenatal clinic tested positive for the disease that year, and S was one of them.³ A fetus does not generally contract HIV in the womb, but an estimated 70,000 infants in South Africa at that time caught the virus each year when exposed to their mother's blood in childbirth or during breastfeeding. In her eighth month of pregnancy, S learned that she had HIV and that her baby was also at risk. S heard on the television that there was a medicine that could stop her baby from getting the disease, but that it was not available in South Africa. S spent two weeks, heavily pregnant and squeezed into hot, crowded minibuses, traveling the dusty, bumpy roads between NGO clinics and faith-based institutions, begging for someone to tell her how she could save her unborn daughter. A tip from a nun led to a hotline, which led to a referral to us. My boss managed to get the correct dose of Nevirapine from a physician friend. Now we just needed to get it to S before she went into labor, which could happen at any moment.